AIRBUS PROBLEM STATEMENTS

AUTONOMOUS SG 2026 | ACACIA HACKATHON SOCIETY X NUS PRODUCT CLUB | UPDATED 31 OCT 2026, 12:45 PM

DATASETS: https://drive.google.com/drive/folders/1w0t4Q_HFT3vzLCwKaq9AYqXGi4O_vhjN

Updates

- The challenges are unchanged; this update adds official clarifications and new data files. **Where the update differs from the original, follow the update.**
- Clarifications are the same on the website, Discord and here.
- Judges accept reasonable assumptions that a team **states clearly** in its demo. A stated assumption is never marked down for being different from ours.
- External public data is allowed if it is cited and fetched by a script. Paid or private data is not.
- Time discipline uses the time of the **first valid submission**. Resubmitting later does not reset it.

Challenge #1

“ How could we leverage AI-driven technologies for satellite collision avoidance? ”

Description

With low Earth orbit (LEO) crowded with satellites and space debris, operators receive thousands of conjunction data messages (CDMs) a day warning of close encounters. Evaluating every encounter by hand takes too long and is vulnerable to human error.

Want to see

A **collision risk evaluator** that reads conjunction telemetry, predicts each event’s final risk tier with a probability, and flags high-risk events that need manoeuvre planning.

Risk classification: in accordance with NASA Spacecraft Conjunction Assessment and Collision Avoidance Best Practices Handbook (NASA/SP-20205011318) and ESA Space Debris Office Standards. $P_c = 10^{risk}$.

Critical (Manoeuvre Required): $P_c \geq 10^{-4}$ (risk ≥ −4.0) or miss distance < 1,000 m

Elevated risk: not Critical, and $10^{-6} \leq P_c < 10^{-4}$ or miss distance 1,000–5,000 m

Safe: $P_c < 10^{-6}$ and miss distance > 5,000 m

Dashboard UI: risk scores and miss distances as closest approach gets nearer, plus a warnings view.

Provided data

Collision Avoidance Challenge dataset (Challenge 1 folder)

- **train_data.csv**: 162,634 CDMs across 13,154 events. **test_data.csv**: 24,484 CDMs across 2,167 events.
- Suggested columns: **event_id**, **time_to_tca**, **miss_distance**, **relative_speed**, **relative_position_r**, **relative_position_t**, **relative_position_n**, **c_object_type**, **c_sigma_r**, **t_sigma_r**, **max_risk_estimate**, **F10**, **AP**. Others are allowed if explained.

- **New:** challenge1_event_labels.xlsx. One row per train event with final risk, final miss distance and final tier (908 Critical, 2,403 Elevated, 9,843 Safe), plus tier rules and data notes.

Clarifications: Challenge #1

- **Units.** **miss_distance** and **relative_position_*** are in **metres**; **relative_speed** is m/s; **time_to_tca** is days. The 1.0 km and 5.0 km thresholds are 1,000 m and 5,000 m.
- **Real task.** Tiers can be read straight from **risk**, so that alone is not the challenge. **Predict the final tier** (from the last CDM before closest approach) using only CDMs with time_to_tca ≥ 2 days.
- **Evaluation.** test_data has no final outcomes; every test CDM is 2–7 days out. Report metrics on a held-out split of train_data **by event_id** (e.g. 80/20). Use test_data to show predictions in the demo.
- **event_id restarts in each file.** Test event 5 is not train event 5. Never join train and test on event_id.
- **risk = −30** (about 41% of rows) is a placeholder for a probability too small to compute. Treat it as $P_c \approx 0$.
- **Class imbalance.** Critical is ~7% of events. Report recall on Critical and a confusion matrix, not only accuracy.
- **Trajectories and 24 hours.** There are no orbits or clock times. Plot relative position (r, t, n) and miss distance against time_to_tca. For warnings, use time_to_tca windows (e.g. closest approach within the next 1 day).

Challenge #2

“ How might AI help air traffic controllers, pilots and ground operations staff make faster and better decisions? ”

Description

Today, when a runway closes, weather delays occur or ground equipment fails, communication relies on phone calls and manual software updates, leading to cascading delays, excessive holding patterns and fuel burn. The future of Air Traffic Management relies on AI assistants that negotiate trade-offs across organisational boundaries.

Want to see

A **Multi-Agent Communication Framework** where ATC, Pilot and Ground Ops AI agents negotiate resolutions to real-time disruptions without human intervention until a consensus is reached.

ATC agent → runway wake separation, taxiway blockage, maximise sector throughput

Pilot agent → preserve fuel reserves, flight safety, minimise turbulence, protect passenger connections

Ground agent → optimise gate use, avoid ground-crew overtime, minimise turnaround delays

Agents exchange structured messages, check cross-domain constraints explicitly, keep a traceable consensus log, and have a deterministic non-AI fallback.

What to avoid

- Combining the 3 roles into a single prompt call.
- Endless, unconstrained agent loops with no convergence logic or turn limit.

- Accepting solutions that ignore safety margins or operational constraints.

Provided data

- US Department of Transportation 2015 flight delays and cancellations (Challenge 2 folder): **airlines.csv**, **airports.csv**.
- **flights.csv was missing and is now added**: flights.csv (358,774 rows, 38.6 MB). It keeps only rows with **WEATHER_DELAY** > 15 or **LATE_AIRCRAFT_DELAY** > 20.
- **New:** challenge2_operational_constraints.xlsx. Gates, runways, runway movements per hour and minimum turnaround per airport; wake separation; fuel reserves; crew duty limits.
- Demonstrate how agents negotiate a **revised arrival time, gate assignment, or flight plan adjustment** within 3 turns.

Clarifications: Challenge #2

- **Input.** Pick any row of flights.csv; that row is the disruption you pass to the 3 agents.
- **One turn** = each of the 3 agents sends one structured message. Reach consensus within **3 turns**. With no agreement after 3 turns, the deterministic fallback decides and the log says why.
- **Missing operational data.** The DOT data has no gates, runways, fuel or crew. Use the new constraints workbook. Its values are **synthetic** planning limits scaled from traffic; you may replace them with cited real figures.
- **Output.** A revised arrival time, gate or flight plan, with the full consensus log.

Challenge #3

“ How might we optimise the SAF supply chain in Kalimantan? ”

Description

There is a need to aggressively scale Sustainable Aviation Fuel (SAF) adoption to meet decarbonisation goals. Kalimantan (Indonesian Borneo) has abundant agricultural and forestry residues, making it a potential hub for regional SAF production. Transporting raw biomass, building infrastructure and delivering SAF to airports requires strategic decisions with complex logistical and capital trade-offs.

Want to see

- A **SAF supply chain optimisation model** that minimises total yearly cost (annualised CapEx + OpEx + logistics) while meeting SAF demand at Kalimantan’s 4 airports.
- Optimal coordinates of depots and refineries, and facility sizing (Small vs Large).
- Material flow network: feedstock to depots, depots to refineries, refineries to airports.

Feedstock available: **6,800 kt of biomass per year**. Capacity is **biomass intake**.

Facility Type	Size	CapEx (USD)	Capacity (kt biomass in / yr)	Fixed OpEx (USD / yr)	Variable OpEx (USD / t in)
Depot	Small	2.25M	150	112,500	5
Depot	Large	4.5M	300	225,000	4
Refinery	Small	480M	500	24.0M	60
Refinery	Large	765M	1,087	38.25M	55

Truck Size	Farm to Depot	Depot to Refinery	Refinery to Airport
Small	3 t	18 t	16 t
Large	9 t	30 t	32 t
Cost (USD / tonne-km)	0.10	0.12	0.08

Airport	IATA	Latitude	Longitude	SAF demand (kt / yr)
Supadio Intl, Pontianak	PNK	-0.15071	109.40400	300
Iskandar, Pangkalan Bun	PKN	-2.70520	111.67300	300
Sultan Aji Muhammad Sulaiman Sepinggan Intl, Balikpapan	BPN	-1.26827	116.89400	300
Juwata Intl, Tarakan	TRK	3.32667	117.56944	300

Data provided

- **Airport locations:** Challenge 3 folder, Airport Location/airports.dat (OpenFlights). Only the four airports above count.
- **Biomass locations:** 80 sites, 20–100 kt each, 6,800 kt total, in Biomass location/kalimantan_80_locations_6800kt_full_range.csv.
- **New:** challenge3_parameters.xlsx (all parameters and assumptions) and challenge3_airports_4.csv.

Clarifications: Challenge #3

- **Capacity means biomass intake** (kt of biomass per year) for both depots and refineries.
- **Yield.** 1 t biomass → 0.20 t SAF. Depots lose no mass (tonnes in = tonnes out).
- **Demand.** 300 kt SAF per airport per year, 1,200 kt in total, so 6,000 kt of biomass is needed. **You do not have to use all 6,800 kt**; choosing which sites to leave out is part of the problem.
- **Time basis: one year.** Annualise CapEx straight-line over 20 years, no interest. Fixed OpEx = 5% of CapEx per year.
- **Distance** = great-circle distance × 1.3 road factor. Truck cost is per tonne-km and the same for both sizes; truck size only changes the number of trips you report.
- **"Iskandar"** means Iskandar Airport, Pangkalan Bun (PKN).
- **Tarakan (TRK) is on an island.** Model the final leg as a barge at the same 0.08 USD per tonne-km, with no extra fixed cost.
- Facilities may sit anywhere on land in Kalimantan. Report their coordinates.

Challenge #4

" How might AI help people find, understand, and verify regulated technical information? "

Description

Sensitive regulatory documents include airworthiness directives, maintenance manuals and accident reports. Mechanics and engineers need precise instructions quickly, with full trust in the output. GenAI models often hallucinate or blend instructions across equipment types, which can cause dangerous mishaps.

Want to see

A **traceable, context-aware search engine** that ingests multi-page technical documents, parses exact applicability rules and answers natural-language queries with citations. If information is ambiguous, it confidently refuses.

Specific references → every claim or sentence links to its exact source metadata

Zero hallucination → the system must not output hallucinated responses

Configuration filtering → evaluate conditional applicability (e.g. Directive A only applies to serial numbers S-1000 to S-1500)

Revision logic → the user knows when a document has multiple versions, and which is latest

Clear uncertainty → out-of-scope, contradictory or ambiguous prompts get an explicit inability message

Sub 3-second retrieval → across 1,000+ pages

Want to avoid

LLM guessing → filling knowledge gaps from pre-trained memory

Unformatted data → returned data must be structured, not loose text

Provided data

- Challenge 4 folder: **AD Notice of Proposed Rulemaking (AD NPRM).xlsx**, **431** FAA airworthiness directive proposals (previously stated as 432).

- **New:** challenge4_ad_index.xlsx. A stable **record_id** per row, ISO dates, **supersedes_ad** extracted from summaries, a serial-number flag, a **revision_chains** sheet and data notes.

Clarifications: Challenge #4

- **They are proposals (NPRMs), not final ADs.** Only 21 rows have a final AD number. Say so in answers where it matters.
- **Revisions.** Every row says Status "Current", so status is not revision data. Use revision_chains: 99 dockets appear more than once, and the latest publish date is the current version. 62 summaries name the AD they supersede.
- **1,000+ pages.** The summaries alone are about 150 pages. Fetch the full documents from the Federal Register links in the sheet (public, no login). This is allowed and encouraged. State your corpus size.
- **Applicability.** Model is filled for only 86 rows and serial numbers appear in only 16 summaries. Read applicability from the full text, and refuse when it is unclear.
- **Citations** = record_id (or docket number) + Federal Register citation and page, plus the section when using full text.

Challenge #5

" How might we leverage historical weather observation logs and flight performance datasets to predict airport runway capacity reductions and flight delay risks 3 to 12 hours before severe weather events occur? "

Description

Severe weather causes most air traffic delays. When it hits major hubs, controllers and operators cut hourly arrival and departure rates only once the weather arrives, so aircraft end up in costly airborne holding or stranded on taxiways, wasting fuel and causing downstream cancellations. Build a **Predictive Airport Capacity & Risk Model** that uses weather observations and flight history to warn 3 to 12 hours ahead.

Want to see

Multi-source data integration → merge weather logs with flight delay records on time and airport

Delay risk classification (main target) → flights facing a severe delay risk (> 45 minutes)

Arrival Acceptance Rate (optional bonus) → hourly AAR forecast, measured by RMSE

Interactive dashboard for ATC → risk scores, historical trends and recommendations

Automatic processing → scripts that download and process public data with no manual steps

Want to avoid

Data leakage across time steps → using weather recorded when the flight arrives to predict its delay

Ignoring runway directionality → raw wind speed without crosswind and tailwind components

Over-fitting to a single airport → the pipeline should accept any airport code

Un-normalised delays → comparing raw delay minutes between hubs and small airports

Provided data

- Challenge 5 folder: **Flight_Data_With_Weather_Final.csv** (148,273 flights, 1 Jan 2023 to 31 Jul 2025). Weather columns **temp**, **prcp**, **wspd**, **wpgt**, **pres**, **snow**, **coco**, each with _ORIGIN and _DEST suffixes.
- Select only **Atlanta Hartsfield-Jackson (ATL)**, all flights at heading **80°**, and **Dallas Fort Worth (DFW)**, all flights at heading **180°** (departing from and arriving to).
- **New:** challenge5_hourly_weather_ATL_DFW.csv: hourly weather for ATL and DFW over the same period **including wind direction**, with headwind, crosswind and tailwind computed. Also challenge5_data_guide.xlsx.

Clarifications: Challenge #5

- **Wind direction was missing**, so crosswind could not be computed. The new hourly file adds it (Meteostat / NOAA ISD stations 72219 ATL and 72259 DFW). Crosswind = speed × |sin(direction − heading)|; headwind = speed × cos(direction − heading); a negative headwind is a tailwind.
- **No leakage.** Weather in the flights file is attached at flight time. Use the hourly file and only weather observed at least **3 hours before** the scheduled time you predict for.
- **Main target** = ARR_DELAY > 45 (arrivals) or DEP_DELAY > 45 (departures) at ATL and DFW. About 10% of flights are positive; report precision and recall, not only accuracy.
- **AAR is optional.** The flights file is a sample (about 6 ATL arrivals a day), so hourly arrival rates cannot be measured from it. Teams who want AAR may download the full BTS on-time data by script.
- **Placeholder columns:** wpgt_* is zero in 98% of rows and snow_* is always zero. Ignore them.
- **Times.** Flight times are local (hhmm). The hourly file gives both UTC and local time.
- **Any airport vs ATL/DFW.** Build the pipeline to take any airport code (IATA or ICAO). Demo and score on ATL and DFW.

END OF PROBLEM STATEMENTS · JUDGING RUBRIC FOLLOWS

Judging Rubric

AI & Aerospace Technology Track

Six criteria. Score each from 1 to 5; apply the weights below. Total weight: 100%.

Problem realism 20%	Technical soundness 25%	Honest evaluation 20%
Trust and traceability 20%	Impact 10%	Time discipline 5%

Problem realism

20% | 1-5

Does this solve a specific aerospace workflow problem?

Name the user, current process and cost in time, money or risk. Explain the constraints of the chosen challenge, rather than describing aviation in general.

STRONG
A credible user, narrow workflow and clear status quo.

WEAK
A generic aviation chatbot or a solution looking for a problem.

Technical soundness

25% | 1-5

Does the prototype work, with the right method for each task?

Show an end-to-end build. Use LLMs for reading and explanation; use solvers, rules or statistics for scheduling, constraints and detection. Test beyond a scripted demo.

STRONG
Working inputs and outputs; explicit constraints and a sound LLM / non-LLM split.

WEAK
LLM-only legality checks, opaque risk scores or a hard-coded demo.

Honest evaluation

20% | 1-5

What was tested, against what baseline, and where does it fail?

State metrics, test data and results. Label synthetic data and its limits. Show one case where the system is wrong, uncertain or declines, and explain how the user knows.

STRONG
A baseline, measured results and a visible failure with a reason.

WEAK
Only success cases, unsupported accuracy claims or hidden data limits.

Judging Rubric

AI & Aerospace Technology Track | Criteria and scoring guidance

Trust and traceability

20% | 1-5

Can a qualified person verify the output before acting?

Trace claims to a document, revision and section, a timestamped signal or an applied rule. Handle applicability and missing evidence. Keep the qualified person in control.

STRONG
Checkable sources, visible uncertainty and a clear refusal when needed.

WEAK
Confident unsourced answers or a claim to replace expert judgement.

Impact

10% | 1-5

What improves over the current process, and by how much?

Compare against the stated baseline: delay minutes, supplier impact, retrieval quality or false alarms, as relevant. Distinguish measured improvement from estimates.

STRONG
A credible before/after measure and realistic adoption constraints.

WEAK
Large benefit claims with no baseline or unstated assumptions.

Time discipline

5% | 1-5

Was the project submitted early and presented efficiently?

Earlier valid submission is better, using the recorded submission time. Also assess a clear, focused presentation completed within 3 minutes. Give one combined 1-5 score.

STRONG
Earlier submission and a focused, complete presentation within 3 minutes.

WEAK
Later submission and a rushed, unfocused or over-time presentation.

SCORING AND JUDGES' NOTES

Scale: 1 = weak; 2 = limited; 3 = credible; 4 = strong; 5 = exceptional evidence.

Weighted total / 100 = sum of (criterion score / 5 x criterion weight).
Example: technical soundness scored 4/5 earns 20 of its 25 points.

Score criteria separately. Ask: "Where does this fail, and how would your user know?"
Reward sound restraint, not presentation polish. Each project is evaluated by 3 judges.
Time is only 5%: earlier submission does not substitute for quality or evidence.